Week1_801_Research Method_Lecture_Note

Research Method 课堂笔记与翻译

Introduction to Research Methods

Lecturer Introduction

My name is Reem Abbas. My background is in computer science. I completed my education up to the master's level in Sudan, followed by a postgraduate diploma in mathematical sciences in South Africa. I then came to New Zealand.

Before arriving, I gained work experience with the United Nations and the International Committee of the Red Cross. I completed my PhD at AUT in Digital Health and Healthcare Informatics within the Department of Computer Science. My research focuses on Disaster Management, such as improving communication during cyclones and earthquakes.

I chair the Disaster Public Health and Healthcare Informatics tracks at the Information System for Crisis Management and Response conference. I am affiliated with Health Informatics New Zealand and am a Fellow of the Australian Institute of Digital Health. I am currently a research associate with AUT and a senior lecturer in technology at UB. I am a mother of three and am originally from Sudan.

Course Context and Research Fundamentals

As evident from my background, my primary interests lie in information systems, digital health, and healthcare informatics. This course is also delivered for business analytics students. While some examples or questions during lectures may be drawn from healthcare, the core principles of research remain consistent across disciplines. The application of research may differ, but the fundamental methodology is universal.

This course serves as an introduction to conducting research, particularly to equip you for the work required in the first trimester. For students without an academic background, applying these skills is crucial, not only for this course but also for successfully submitting your assessments.

Expectations for Self-Directed Learning and Academic Writing

If you recall from the orientation, I asked you to review a scientific paper. As Arun mentioned, the course structure involves 25% face-to-face communication and 75% self-directed learning. When submitting assessments, you are expected to write in an academic style.

Reviewing scientific papers will help you understand the expected standard. Another useful practice is to type any paragraph or piece of work you intend to submit and use online tools to compare it against academic writing standards.

Academic writing means we cannot use casual language. Expressions such as "I think" or "this is because" are generally not appropriate for formal assessments. Every piece of information or claim in your assessments must be supported by references.

> 🧑‍🏫 Lecturer: I'll give you an example. If I'm writing about disasters and state, "disasters are increasing in frequency," I must either cite an academic source for this claim or present it as a conclusion from my own analysis of data.

研究方法导论

讲师介绍

我是Reem Abbas。我的背景是计算机科学。我在苏丹完成了直到硕士阶段的教育，随后在南非获得了数学科学的研究生文凭。之后我来到了新西兰。

在抵达新西兰之前，我在联合国和红十字国际委员会获得了工作经验。我在奥克兰理工大学（AUT）计算机科学系完成了数字健康与医疗信息学的博士学位。我的研究专注于灾害管理，例如在气旋和地震期间改善通信。

我在危机管理与响应信息系统会议上担任灾害公共卫生与医疗信息学分论坛的主席。我是新西兰健康信息学的成员，也是澳大利亚数字健康研究所的研究员。我目前是AUT的研究助理，也是UB（注：可能指另一所大学）的技术高级讲师。我是三个孩子的母亲，原籍苏丹。

课程背景与研究基础

从我的背景可以看出，我的主要兴趣在于信息系统、数字健康和医疗信息学。这门课程也面向商业分析专业的学生授课。虽然讲座中的一些例子或问题可能来自医疗保健领域，但研究的核心原则在各个学科中是一致的。研究的应用可能有所不同，但基本的方法是普适的。

本课程旨在介绍如何进行研究，特别是为你们在第一个学期所需的工作做好准备。对于没有学术背景的学生来说，应用这些技能至关重要，不仅是为了这门课程，也是为了成功提交你们的评估作业。

对自主学习和学术写作的期望

如果你们还记得在迎新会上，我要求你们查阅一篇科学论文。正如Arun所提到的，课程结构包括25%的面对面交流和75%的自主学习。在提交评估作业时，你们需要以学术风格进行写作。

查阅科学论文将帮助你们理解预期的标准。另一个有用的做法是，将任何打算提交的段落或作品输入电脑，并使用在线工具将其与学术写作标准进行比较。

学术写作意味着我们不能使用随意的语言。诸如“我认为”或“这是因为”之类的表达通常不适合正式的评估作业。你们评估作业中的每一条信息或主张都必须有参考文献支持。

> 🧑‍🏫 讲师：我举个例子。如果我在写关于灾害的文章，并陈述“灾害发生的频率正在增加”，那么我必须为此主张引用学术来源，或者将其作为我自己分析数据后得出的结论来呈现。

Academic Writing and Research Fundamentals

Core Principles of Academic Writing

In formal assessments, we cannot include casual expressions like "so this is because". Every piece of information or claim in your assessments **must be backed by references**.

> 🧑‍🏫 Lecturer: I'll give you an example. If I'm writing about disasters and state "disasters are increasing in frequency," I must either cite an academic source for this claim or present it as a conclusion from my own research analysis.

Therefore, any such statements must be supported by a source. Crucially, it must be an **accredited scientific source**. We will cover what this means, including which sources are reliable, where to find good papers, and which websites to avoid. In general, sources must be **peer-reviewed**.

Peer-reviewed means the paper has been scientifically validated or reviewed by field experts—reputable individuals who can confirm it is credible information suitable for use in your research.

Course Overview and Research Foundations

Before we begin, some housekeeping. The course content is found in your program document. Today we will cover:

1. Introduction to research.
2. Fundamentals of research.
3. Most importantly, **how to develop research questions**.

Any research starts with a question. Research means trying to answer a question without bias or personal opinions; it's about answering questions **scientifically**. We do this scientifically to **avoid bias**.

The first assessment will focus on developing good research questions. **Everything in research starts from and aims to answer a research question**.

The Research Process

Once you have a question, the next step is to design your study. When you define or develop your question, you must understand exactly what you want to do.

- Your question must be **specific**.
- Your question must be **feasible**—answerable within the time and resources you have.

After defining the question, you need to design your study. This involves planning how you will collect data and how you will analyze it. We will then discuss different types of methods, such as **qualitative or quantitative** methodologies.

Research in the New Zealand Context

We will also talk about research methodologies when dealing with Māori and Pacifica communities. This is critically important because we live in New Zealand, a country founded on a document called the **Treaty of Waitangi**.

> 🧑‍🏫 Lecturer: For international students or those unfamiliar with the context: Māori are the indigenous people of New Zealand. The country was colonized by Europeans, primarily the British. The foundation for the peaceful coexistence of these two nations was signed in what we know as the Treaty of Waitangi.

Understanding this is vital, not only for research but also for future job interviews, where you will likely be asked about the **principles of the Treaty of Waitangi**. It is fundamental for anyone living in New Zealand to understand these three principles, as they influence everything, from research to broader study and societal engagement.

学术写作与研究基础

学术写作核心原则

在正式的评估作业中，我们不能包含诸如“所以这是因为”之类的随意表达。你评估作业中的每一条信息或主张**都必须有参考文献支持**。

> 🧑‍🏫 讲师：我举个例子。如果我在写关于灾害的文章，并陈述“灾害发生的频率正在增加”，那么我必须为此主张引用学术来源，或者将其作为我自己分析数据后得出的结论来呈现。

因此，任何此类陈述都必须有来源支持。至关重要的是，它必须是一个**经认证的科学来源**。我们将探讨这意味着什么，包括哪些来源是可靠的、在哪里可以找到好的论文，以及应该避免哪些网站。一般来说，来源必须是**经过同行评审的**。

同行评审意味着该论文已经过科学验证或由该领域的专家——能够确认其是适合用于你研究的可信信息的权威人士——审阅过。

课程概述与研究基础

在开始之前，是一些行政事务。课程内容在你们的课程文件中。今天我们将涵盖：

1. 研究导论。
2. 研究基础。
3. 最重要的是，**如何提出研究问题**。

任何研究都始于一个问题。研究意味着试图不带偏见或个人观点地回答一个问题；它是关于**科学地**回答问题。我们科学地进行研究是为了**避免偏见**。

第一次评估将侧重于提出好的研究问题。**研究中的一切都始于并旨在回答一个研究问题**。

研究过程

一旦你有了问题，下一步就是设计你的研究。当你定义或提出问题时，你必须确切地理解你想要做什么。

- 你的问题必须**具体**。
- 你的问题必须**可行**——在你拥有的时间和资源范围内能够回答。

在定义问题之后，你需要设计你的研究。这包括规划你将如何收集数据以及你将如何分析数据。然后我们将讨论不同类型的方法，例如**定性或定量**方法论。

新西兰背景下的研究

我们还将讨论在与毛利人和太平洋岛民社区打交道时的研究方法论。这一点至关重要，因为我们生活在新西兰，一个建立在名为**《怀唐伊条约》** 的文件之上的国家。

> 🧑‍🏫 讲师：对于国际学生或不熟悉此背景的同学：毛利人是新西兰的原住民。这个国家曾被欧洲人（主要是英国人）殖民。这两个民族在新西兰和平共处的基础就是我们所知的《怀唐伊条约》。

理解这一点至关重要，不仅对于研究，对于未来的工作面试也是如此，面试中你很可能会被问到**《怀唐伊条约》的原则**。对于任何生活在新西兰的人来说，理解《怀唐伊条约》的这三项原则是根本，因为它们影响着从研究到更广泛的学习和社会参与的一切事务。

Treaty of Waitangi and Research Implications

The peaceful coexistence of the two nations in New Zealand is based on the **Treaty of Waitangi**. It is fundamental for anyone living in New Zealand to understand its **three principles**, as they influence everything from research and study to work interviews.

Therefore, in your free time, please go and understand: why was it signed, what does it entail, and what is the Treaty of Waitangi? Its implication for research methods is that we must take the Treaty into consideration. This means we must be **culturally sensitive** to the Māori population, the indigenous people and owners of the country.

We may invite guest lecturers, including a **Māori researcher**, to explain how conducting research with Māori differs from other populations. In every course, remember that **cultural awareness** is crucial, especially in fields like healthcare where understanding cultural sensitivity with Indigenous populations is vital.

> 🧑‍🏫 Lecturer: Does that make sense? Any questions? Please feel free to jump in. Don't wait. Let's make it interactive. If you have examples or experiences from your background

or career, feel free to share them.

Course Communication and Structure

We will use **Microsoft Teams** for our communication. You can contact us in two ways: via **Teams** or **email**.

- **Student Email:** `yourstudentID@ubstudent.ac.nz`
- **Lecturer Email:** `firstname.lastname@ubcolleges.com`

For important matters that need documentation—like signing or sending documents—or for serious issues to discuss, please use **email**. For questions, accessing session recordings, and casual conversation with peers, use **Teams**.

Session Recordings

We usually record sessions. However, as our communication is face-to-face this time, I will not record this session. If you wish to review content, I can post recordings from the last term.

Issue Resolution Pathway

For general communication issues, please follow this pathway:

1. First, go to your tutor (Mohammed).
2. You can also have a class representative to speak on behalf of the class.
3. For personal issues, go directly to your Teaching Assistant (TA).
4. If unresolved by your TA, escalate to your team leader, Dr. Arun Kumar.
5. If, for any reason, it remains unresolved (which we haven't experienced yet)...

《怀唐伊条约》与研究意义

新西兰两个民族的和平共处建立在**《怀唐伊条约》** 的基础之上。对于任何生活在新西兰的人来说，理解其**三项原则**是根本，因为它们影响着从研究、学习到工作面试等方方面面。

因此，请在空闲时间了解：它为何被签署、包含什么内容、以及《怀唐伊条约》究竟是什么？它对研究方法的意义在于，我们必须将条约纳入考量。这意味着我们必须对毛利人群体——这个国家的原住民和所有者——保持**文化敏感性**。

我们可能会邀请客座讲师，包括一位**毛利研究员**，来解释与毛利人进行研究与其他人群有何不同。在每门课程中，请记住**文化意识**至关重要，尤其是在医疗保健等领域，理解与土著居民相处的文化敏感性至关重要。

> 🧑‍🏫 讲师：明白了吗？有什么问题吗？请随时提问。不要等待。让我们互动起来。如果你有来自个人背景或职业经历的案例或经验，请随时分享。

课程沟通与结构

我们将使用 **Microsoft Teams** 进行沟通。你可以通过两种方式联系我们：通过 **Teams** 或**电子邮件**。

- **学生邮箱：** `你的学号@ubstudent.ac.nz`
- **讲师邮箱：** `名.姓@ubcolleges.com`

对于需要记录的重要事项——例如签署或发送文件——或需要讨论的严肃问题，请使用**电子邮件**。对于提问、获取课程录制内容以及与同学进行随意交流，请使用 **Teams**。

课程录制

我们通常会录制课程。然而，由于本次是面对面交流，本次课程将不进行录制。如果你想复习内容，我可以发布上学期的录制内容。

问题解决途径

对于一般的沟通问题，请遵循以下途径：

1. 首先，联系你的导师（Mohammed）。
2. 你也可以选出一位班级代表为全班发言。
3. 对于个人问题，直接联系你的助教（TA）。
4. 如果助教无法解决，请升级到你的团队负责人，Arun Kumar 博士。
5. 如果出于任何原因问题仍未解决（这种情况我们尚未遇到过）...

内容，我也可以发布。

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5. 如果出于任何原因问题仍未解决（这种情况我们尚未遇到过），Arun Kumar 博士是一位非常支持员工的领导，那么问题将升级到校区经理。
6. 之后，问题会升级到学院等，但我认为没有人需要走到那一步。

沟通规范

- 在 Teams 上给我发消息，请在 **上午9点至下午5点** 之间。
- 我会在这些时间段内或看到消息时回复。
- 我们周四和周五休息。
- 对于重要或紧急事务，请使用电子邮件。

课程结构

通常的课程结构如下（今天除外，今天只是介绍和概述）：

1. **前半部分**：讲授主要内容。
2. **课间休息**：大约15分钟。
3. **后半部分**：进行小组练习，以确保理解材料。

这门课程是**理论性**的，与你将学习的其他课程完全不同。对于没有学术背景的人来说可能有些挑战。如果你感到困惑，请随时提问，我随时为你提供支持。

学习资源

所有课程资料都将在 Blackboard 上的 **MSE 801 研究方法** 频道中提供。你将找到：

- 幻灯片
- 供你本周练习的**自主学习**材料

- 如果你想深入探讨某个主题，还有**延伸阅读材料**

出勤与守时

- 休息时间为 **10到15分钟**。请休息后准时返回。迟到会干扰讲师，你也会错过重要内容。
- 如果迟到超过15分钟，将被视为**缺勤**。
- 如果你因任何情况无法出席，请**提前**与我联系，你的缺勤将被视为合理，不会影响你的出勤率或移民要求。
- 如果你不提前说明，缺勤将被视为不合理，并对你产生负面影响。
- 你需要保持 **90%** 的出勤率。

课程评估

本学期的**最后一节课**（或最后两节课，因为这是一个大班）将用于学生**展示**。这通常是评估任务之一，目的是确保你理解所提交的内容，并提升你的演讲技巧。

内容，如有需要，我也可以发布。

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- 如果你因任何情况无法出席，请**提前**与我联系，你的缺勤将被视为合理，不会影响你的出勤率或移民要求。
- 如果你不提前说明，缺勤将被视为不合理，并对你产生负面影响。
- 你需要保持 **90%** 的出勤率。

课程评估

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课程安排与评估详情

由于这是一个大班，我们可能需要安排**最后两节课**用于学生**展示**。这通常是评估任务之一，目的是确保你理解所提交的内容，并提升你的演讲技巧。

学习资源与客座讲座

- 所有学习资源、自主学习的延伸阅读材料以及评估详情都将发布在**Blackboard**上。
- 我们偶尔会安排**客座讲座**，通常为在线形式，时长约一小时。
- 我会提前告知讲座主题，请大家积极思考并互动提问。

评估结构与时间

本课程有两项评估：

1. **第一项评估**占总成绩的 **40%**。
2. **第二项评估**占总成绩的 **60%**。

评估为**开卷**形式。每项评估发布后，你有**三周**时间完成并提交。成绩将在截止日期后的三周内公布。成绩由多位老师共同批阅。

> 🧑‍🏫 讲师：如果你对成绩有疑问，请在成绩公布后的**大约两天内**提出，我们会进行核查。请不要拖延，因为成绩公布后还有其他流程需要跟进。

学术写作与诚信

学术写作和正确引用至关重要。请务必使用正确的引用格式。

- 在UBI，我们使用 **APA第7版** 格式。相关教学视频可在自主学习材料中找到。
- 在学术写作中，必须**坚持使用一种**引用格式，并且不要引用无意义的网络链接。
- **最重要的是学术诚信**。评估虽然是开卷，但请不要冒险导致课程不及格。

> 🧑‍🏫 讲师：例如，如果一项评估要求你选择并评述科学论文，而你却将问题输入ChatGPT等工具，它可能会生成虚假的论文。我们甚至不会继续看你写了什么，这直接导致**不及格**。如果连论文都不是学生自己选的，这也是明确的**不及格**。即使重交，成绩也会被限制在50分（即C-）。所以，请不要涉足这个领域。你可以用它来辅助校对，但不要用它来获取答案。

Course Schedule and Assessment Details

Because this is a large cohort, we may need to schedule the **last two sessions** for student **presentations**. This is usually one of the assessment tasks, aiming to ensure you understand what you have submitted and to improve your presentation skills.

Learning Resources and Guest Lectures

- All learning resources, self-directed learning materials, and assessment details will be on **Blackboard**.
- We occasionally have **guest lectures**, typically online and lasting about one hour.
- I will inform you of the topic beforehand. Please think about it, be interactive, and ask questions.

Assessment Structure and Timeline

There are two assessments for this course:

1. **The first assessment** is worth **40%** of the total grade.
2. **The second assessment** is worth **60%** of the total grade.

Assessments are **open book**. You have **three weeks** to complete and submit each one after its release. Marks will be released three weeks after the deadline. Your work will be marked by multiple people.

> 👨 🏫 Lecturer: If you have any concerns about your marks, you have **around two days** after their release to raise them, and we will look into it. Please don't take longer, as other processes follow after we release the marks.

Academic Writing and Integrity

Academic writing and correct referencing are crucial. Please use the right reference format.

- At UBI, we use the **APA 7th edition** format. Videos on how to do this are available in the self-directed learning materials.
- In academic writing, you must **stick to one** formatting style and not include nonsensical website links.
- **Most importantly, academic integrity**. Although assessments are open book, please don't risk failing the course.

> 👨 🏫 Lecturer: For example, if an assessment asks you to choose and review scientific papers, and you put the questions into ChatGPT or similar tools which might generate fake papers, we won't even look further at what you've written. That's a reason for **failure**.

> If the paper wasn't even chosen by the student, that's a straightforward **failure**. When you resubmit, your mark is capped at 50 (a C-). So, don't go to that area. You can use it for proofreading, but not for expecting answers.

Academic Integrity and Its Importance

Using AI to generate papers and submitting them is a reason for **failure**. If the paper is not even chosen by the student, that's a straightforward **failure**. When you resubmit, your mark is capped at 50 (a C-). So, don't go to that area. You can use it for proofreading, but not for expecting answers and then submitting them.

Attendance should be 90%. If you have any circumstances, send them to me and it will be justified, so it will not impact your record. For other cohorts, they have TAs, but for the MSE, I will be your only person delivering the course.

What is Academic Integrity?

Before asking you not to violate it, I must be clear about what is possible and what is not allowed. **Academic Integrity** refers to maintaining honesty and fairness in all academic work and upholding ethical standards in research.

This means:

- Do not copy from someone.
- Do not pay someone to do your work for you.
- Do not falsify data.

Academic integrity is not only about asking for help. For example, when you read a paper, the information you get should be accurately represented in your assessments. Do not change or falsify data, information, or results in your research. Unauthorized collaboration with others on assignments or projects is also prohibited. These are all things we should avoid.

Plagiarism and Proper Citation

Plagiarism is the most common form of academic integrity violation, which is using someone else's work or ideas without proper citation. This is the main reason why we use references.

When you write a paper or an assessment, you will definitely use other people's work. That's totally fine because we don't know everything. However, you must give them credit. If you write something uncited, it implies you are the owner of that information, like a copyright. For important information, statistics, or credible data, you must cite the person or source from which you obtained it.

Key Principles

To uphold academic integrity:

1. **Cite your sources.** Always acknowledge where your information comes from.
2. **Do your own work.** Complete your assignments and exams independently (unless it's a group assignment, which we don't have in Research Methods).
3. **Seek help appropriately.** Ask tutors and do not seek unauthorized help.

Don't risk failing this course. This is just a quick overview of academic integrity.

> 🧑‍🏫 Lecturer: Please let me know. Give me a thumbs up if you can hear me.

The hardest thing is to get started. You have to get it on paper. Thank you.

Student Perspectives on Integrity

> 🙋‍♂️ Student 1: Integrity... I think the word *ethics* kind of comes to my mind. 🙋‍♂️ Student 2: It's a personal responsibility that we all share. 🙋‍♂️ Student 3: It means honesty to me. 🙋‍♂️ Student 4: I think of academic integrity as creating your own work. 🙋‍♂️ Student 5: It is important to submit work of your own because it shows the lecturer you know what you're talking about. 🙋‍♂️ Student 6: You can work with others and rely on your peers, but to create something that's genuine and your own... because at the end of the day, it is something that you submit as your own work, just like referencing when they're doing their work.

学术诚信及其重要性

使用人工智能生成论文并提交，是导致**不及格**的一个原因。如果论文甚至不是由学生选择的，那就是明确的**不及格**。当你重新提交时，你的分数将被限制在50分（C-）。所以，不要涉足那个领域。你可以用它来校对，但不能指望它提供答案然后提交。

出勤率应达到90%。如果你有任何特殊情况，请发送给我，这将被合理解释，因此不会影响你的记录。对于其他班级，他们有助教，但对于软件工程硕士（MSE）课程，我将是唯一授课的人。

什么是学术诚信?

在要求你们不要违反之前，我必须明确什么是允许的，什么是不允许的。**学术诚信**是指在所有学术工作中保持诚实和公平，并在研究中维护道德标准。

这意味着：

- 不要抄袭他人。
- 不要付钱请人为你做作业。
- 不要伪造数据。

学术诚信不仅关乎寻求帮助。例如，当你阅读论文时，你获得的信息应该在评估中准确呈现。不要在你的研究中更改或伪造数据、信息或结果。未经授权在作业或项目中与他人合作也是被禁止的。这些都是我们应该避免的事情。

抄袭与正确引用

抄袭是最常见的学术诚信违规形式，即未经适当引用而使用他人的作品或观点。这是我们使用参考文献的主要原因。

当你撰写论文或进行评估时，你肯定会使用他人的作品。这完全没问题，因为我们不可能无所不知。然而，你必须给予他们应有的荣誉。如果你写了未引用的内容，就意味着你是该信息的所有者，类似于拥有版权。对于重要信息、统计数据或可信数据，你必须注明获取该信息的个人或来源。

关键原则

为维护学术诚信：

1. **引用你的来源。** 始终注明你的信息来源。
2. **独立完成作业。** 独立完成你的作业和考试（除非是小组作业，而研究方法课中没有小组作业）。
3. **以适当方式寻求帮助。** 向导师请教，不要寻求未经授权的帮助。

不要冒不及格的风险。这只是对学术诚信的简要概述。

> 🧑‍🏫 讲师：请让我知道。如果你能听到，请给我点个赞。

最难的是开始。你必须把它落实到纸上。谢谢。

学生对诚信的看法

> 🙋 学生1：诚信……我想到了*道德*这个词。 🙋 学生2：这是我们所有人共同承担的个人责任。 🙋 学生3：对我来说，它意味着诚实。 🙋 学生4：我认为学术诚信就是创作自己的作品。 🙋 学生5：提交自己的作品很重要，因为这向讲师表明你知道自己在说什么。 🙋 学生6：你可以与他人合作并依赖同伴，但要创造出真实且属于自己的东西……因为归根结底，这是你作为自己的作品提交的东西，就像他们在做自己的工作时需要引用一样。

> 🙋 学生4：我认为学术诚信就是创作自己的作品。提交自己的作品很重要，因为这向讲师表明你知道自己在说什么。你可以与他人合作并依赖同伴，但要创造出真实且属于自己的东西……因为归根结底，这是你作为自己的作品提交的东西，就像他们在做自己的工作时需要引用一样。

> 🙋 学生5：我经历的所有学习……对我来说都是一个学习过程，比如使用 Turnitin 这类软件来检查我是否抄袭了他人的作品。

> 🙋 学生6：就我个人而言，我觉得这很有意义。作为一名学生，我认为秉持学术诚信能让我以最好的方式完成学业。这基本上就是在展示我的诚实。我所做的一切都源于诚实。但在分享方面，重要的是完成自己的工作。我认为，当你创作出与其他人略有不同的作品时，这真的很有成就感。

> 🙋 学生7：它确实会渗透到你做的每一件小事中。在学生生涯中以诚实和真实的方式行事，最终将帮助你达到未来想要达到的目标。

> 🙋 学生8：说实话，一个没有学术诚信的世界将是不公平的。那将是一场大战。每个人的研究都会几乎相同。将会有很多并不真正配得上的人占据职业阶梯的高位。而且，我觉得这只会让世界变得一团糟。

> 🙋 学生9：我们可能正在接近那样一个世界，那里有很多……有很多精明的数字途径，确实可能跨越学术诚信的界限。这个世界已经有太多不公平，我们真的不想再增加了。我觉得每个人都会在一层谎言之上再堆一层，我们将永远无法知道真相。因此，我相信学术诚信对于社会的发展以及个人真正到达他们应得的位置都至关重要。

Session Introduction and Learning Objectives

好的。以上就是引言部分，设定了场景，让大家了解了要求和期望。那么，在今天这堂课中，我们将讨论**研究的含义**。这包括其目的、动机、研究类型、方法类型以及**科学方法**。

在研究的每一个方面，我们都必须讨论**伦理**。我们如何在这个过程中应用伦理？

学习目标

请大家花几秒钟时间看一下这些学习目标，以便我们能达成共识。

The Meaning of Research

那么，研究的含义是什么？我希望你们回想一下，当你需要为个人或职业决策研究某件事的时候。例如，你决定在 UBI 攻读 MBA 或 MSE。你采取了哪些步骤？你是如何收集信息的？

任何决策，在做出之前，你都会问自己一些问题：这对我合适吗？这是正确的步骤吗？我为什么要读这个 MSE？我为什么要读这个硕士？

如果我询问某人攻读 MSE 或 MBA 的经验，无论是什么项目，答案都源于他们自身的需求。对我来说，在我人生的这个阶段攻读硕士可能是正确的决定。但对其他人来说，这可能就不合适。

因此，为了避免偏见，也为了避免回答仅反映个人观点的问题，我们需要进行**系统性的**研究。

研究的含义与学习目标

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因此，为了避免偏见，也为了避免回答仅反映个人观点的问题，我们需要进行**系统性的**研究。

Research: A Systematic Approach to Answering Questions

Any decision, before making it, you ask yourself some questions: Is this right for me? Is this the right step to take? Why am I doing this MSE? Why am I doing this Master's?

If I ask someone about their experience doing an MSE or an MBA, whatever the program is, the answer comes from their own needs. For me, it could be the right decision to take a Master's at this stage in my life. For somebody else, it might not be the right thing.

So, to avoid bias and to avoid answering a question that reflects an individual perspective, we need to do this **systematically**, **scientifically**. We need to have a scientific way of answering our questions.

The Research Process

It all starts with a question. In this case, it would be: "Is it the right time and place for me to do a Master's or not?" For this, you have to collect information. You need to know something about New Zealand, why New Zealand, and in New Zealand, why UB? Why not the University of Auckland or any other institution? What are the benefits? What is the potential for achieving something else? That's the information you need to collect.

Now, after collecting so much information, you need to analyze it to reach a final decision. When you do this systematically, you are **researching**.

The Core Steps of Research

1. Research starts with a **question**.
2. Then you **collect data** to answer that question.
3. You have a lot of information, so you need to **analyze** it to answer the question.

Through **systematic methods**, research allows us to see things more clearly and objectively, free from bias. If it wasn't systematic and objective, then one person's experience would be different from another's, and so on.

Defining Research

A comprehensive definition of research is: "The manipulation of things, concepts or symbols for the purpose of generalizing to extend, correct or verify knowledge, whether that knowledge aids in construction of theory or in practice of art." This is quite complicated.

In simple words, to do research, you can't ask *everyone* who did a Master's at UB for information. You have to choose a **limited number of people** (a sample), collect information from them, and then **generalize** the findings.

Example: Generalizing from a Sample

For example, if I'm trying to understand the average age at which people graduate from university in New Zealand, China, or Egypt, I cannot ask everyone. I have to **sample**. I have to have a **sample**, analyze the data collected from that sample, and then I can **generalize**. All these steps are done systematically using tools like **statistics** and **descriptive analysis**.

Research vs. Engineering Projects

Research involves an **original contribution to the existing stock of knowledge**. How do we explain this? You are in the Master of Software Engineering, building a platform, an app, a website, or an information system.

What makes this *research* versus just *building a system*? If you're building a system and just using it within its intended stakeholders, this is simply an **engineering project**. However, when you **share the value or the benefits** of this system with the wider community, it becomes **research**. Sharing requires that you **evaluate the benefit** of what you have created. So, there's a difference between just building something and conducting research about it.

研究：回答问题的系统性方法

任何决策，在做出之前，你都会问自己一些问题：这对我合适吗？这是正确的步骤吗？我为什么要读这个 MSE？我为什么要读这个硕士？

如果我询问某人攻读 MSE 或 MBA 的经验，无论是什么项目，答案都源于他们自身的需求。对我来说，在我人生的这个阶段攻读硕士可能是正确的决定。但对其他人来说，这可能就不合适。

因此，为了避免偏见，也为了避免回答仅反映个人观点的问题，我们需要**系统性地**、**科学地**进行。我们需要一种科学的方法来回答我们的问题。

研究过程

一切都始于一个问题。在这个例子中，问题可能是："现在是否是攻读硕士的正确时间和地点？"为此，你必须收集信息。你需要了解新西兰的情况，为什么选择新西兰，以及在新西兰，为什么选择 UB？为什么不选择奥克兰大学或其他机构？有什么好处？实现其他目标的潜力是什么？这些就是你需要收集的信息。

现在，收集了大量信息后，你需要进行分析以做出最终决定。当你系统性地做这件事时，你就是在进行**研究**。

研究的核心步骤

1. 研究始于一个**问题**。
2. 然后你**收集数据**来回答这个问题。
3. 你拥有大量信息，因此需要**分析**这些信息来回答问题。

通过**系统性方法**，研究使我们能够更清晰、更客观地看待事物，避免偏见。如果它不是系统性和客观的，那么一个人的经验就会与另一个人的不同，依此类推。

定义研究

研究的一个全面定义是："为了概括、扩展、纠正或验证知识而对事物、概念或符号进行的操作，无论该知识是用于构建理论还是实践艺术。"这相当复杂。

简单来说，做研究时，你不能询问*每*一个在 UB 读过硕士的人。你必须选择**有限数量的人**（一个样本），从他们那里收集信息，然后**概括**研究结果。

示例：从样本中概括

例如，如果我想了解在新西兰、中国或埃及，人们从大学毕业的平均年龄，我不能询问所有人。我必须**抽样**。我必须有一个**样本**，分析从该样本收集的数据，然后我才能**概括**。所有这些步骤都是系统性地完成的，使用**统计学**和**描述性分析**等工具。

研究与工程项目

研究涉及**对现有知识体系的原创性贡献**。如何解释这一点？你正在攻读软件工程硕士，构建一个平台、应用程序、网站或信息系统。

是什么让这成为*研究*，而不仅仅是*构建系统*？如果你构建一个系统，并且仅在其目标利益相关者内部使用，这只是一个**工程项目**。然而，当你与更广泛的社区**分享**该系统的**价值或益处**时，它就变成了**研究**。分享要求你**评估**你所创造之物的**益处**。因此，仅仅构建某物与对其进行研究是有区别的。

What Makes It Research vs. Just Building?

What makes this **research** versus just building a system? If you're building a system and using it only within its intended stakeholders, this is simply an **engineering project**. However, when you **share** the **value or benefits** of this system with the wider community, it becomes **research**. Sharing requires you to **evaluate** the **benefit** of what you have created. Therefore, there is a difference between merely building something and ensuring it constitutes research.

By the end of this course, you will understand this difference and be prepared to conduct research. Ultimately, the key takeaway is that **research adds to the existing stock of knowledge**. You conduct your research, publish it, and it becomes part of the collective knowledge base, adding to what people know.

The Purpose and Impact of Research

The purposes of research are multifaceted:

1. **Evidence-Based Decision-Making:** We need scientific ways to answer questions and avoid biased feedback.
2. **Knowledge Generation & Innovation:** Research helps generate new knowledge or innovate, such as applying AI in healthcare or business for efficiency.

3. **Improving Lives:** For example, researching the impact of old houses in New Zealand on healthcare helps improve living conditions.
4. **Answering Challenging Questions:** It solves real-life issues.
5. **Creating Opportunities:** Research can open doors to new markets or identify viable business opportunities.
6. **Establishing Partnerships:** It facilitates collaborations, like a university partnering with a college on an investigative project.
7. **Societal Impact & Economic Boost:** Research benefits the community and is crucial for turning knowledge into products/services, thereby boosting economies.

Before any innovation or new venture—like starting a business in New Zealand—you must conduct **evidence-based** research. This research should be grounded in statistics, economic data, social factors, potential client base, and success rates.

Key Characteristics of Research

Before delving into details, consider what should be present in a research project. You need to be **curious** and genuinely want to know about your topic. Research involves answering questions and checking the **consistency and quality** of your steps.

> 🙋 Student: At this stage, maybe some of you would find this very theoretical or not. You can't link this to the practical steps of doing research. 👨‍🏫 Lecturer: But don't worry. In the coming sessions, as we collect data, analyze data, and answer questions, these concepts will become clearer.

A fundamental characteristic is that **research has to use precise data**. For instance, if collecting data to understand the number of students undergoing stress at UB, the data must be accurate and methodically gathered.

是什么让它成为研究而不仅仅是构建?

是什么让这成为**研究**，而不仅仅是构建一个系统？如果你构建一个系统，并且仅在其目标利益相关者内部使用，这只是一个**工程项目**。然而，当你与更广泛的社区**分享**该系统的**价值或益处**时，它就变成了**研究**。分享要求你**评估**你所创造之物的**益处**。因此，仅仅构建某物与确保其构成研究是有区别的。

在本课程结束时，你将理解这种差异，并为进行研究做好准备。最终，关键要点是：**研究增加了现有的知识储备**。你进行研究，发表成果，它就成为集体知识库的一部分，增加了人们所知的内容。

研究的目的与影响

研究的目的多方面：

1. **基于证据的决策：** 我们需要科学的方法来回答问题，避免有偏见的反馈。
2. **知识生成与创新：** 研究有助于生成新知识或创新，例如将人工智能应用于医疗保健或商业以提高效率。
3. **改善生活：** 例如，研究新西兰老房子对医疗保健的影响有助于改善生活条件。
4. **回答具有挑战性的问题：** 它解决现实生活中的问题。
5. **创造机会：** 研究可以打开新市场的大门或识别可行的商业机会。
6. **建立伙伴关系：** 它促进合作，例如大学与学院合作进行调查研究项目。
7. **社会影响与经济促进：** 研究使社区受益，并且对于将知识转化为产品/服务、从而促进经济至关重要。

在任何创新或新事业之前——比如在新西兰创业——你必须进行**基于证据的**研究。这项研究应基于统计数据、经济数据、社会因素、潜在客户群和成功率。

研究的关键特征

在深入细节之前，请思考一个研究项目应具备什么。你需要有**好奇心**，并且真正想了解你的主题。研究涉及回答问题并检查你步骤的**一致性和质量**。

> 🙋 学生：在这个阶段，也许你们中的一些人会觉得这非常理论化，或者无法将其与研究的具体步骤联系起来。 🧑‍🏫 讲师：但别担心。在接下来的课程中，随着我们收集数据、分析数据和回答问题，这些概念会变得更加清晰。

一个基本特征是：**研究必须使用精确的数据**。例如，如果收集数据以了解UB大学承受压力的学生数量，数据必须准确且有方法地收集。

Research Characteristics: Data Integrity and Ethical Conduct

At this stage, some of you might find this very theoretical or struggle to link it to the practical steps of doing research. But don't worry. In the coming sessions, as we collect data, analyze data, and answer questions, these concepts will become clearer.

A fundamental characteristic is that **research must use precise data**. For example, if collecting data to understand the number of students undergoing stress at UB, the data must be accurate. If you don't use accurate data, your findings will be inaccurate, leading to flawed decisions based on nothing. Therefore, data gathering and verification must be precise.

If you are using data from other sources, you need to ensure or verify that this data is reliable. In your assessments, you achieve this by choosing **peer-reviewed publications**. You must also maintain **trustworthiness** and be careful about **integrity**.

The Importance of Ethical Integrity: A Case Study

I'll give you an example. When conducting research, you'll find that indigenous populations, like Māori communities, are very strict about sharing information. You cannot approach a Māori community directly for research. You must go through what we call a **Māori consultant**—a trusted person who knows you and the community—to establish trust before conducting interviews or collecting data.

This requirement stems from very valid historical reasons. Historically, there have been abuses of these communities. Data collected about them has been published in ways that do not reflect what they actually said.

For instance, consider the statement that obesity is higher in Māori and Pacific communities. A simplistic, unethical conclusion might be that they are not following healthy lifestyles or not exercising enough. This is unscientific and dishonest. Proper research is needed to understand the reasons, which could be:

- Genetic factors.
- Stress.
- Socioeconomic status (e.g., inability to afford healthy food versus cheaper fast food).

When you take data, analyze it, and then fail to represent it correctly, this demonstrates a **lack of honesty and lack of integrity**. As a researcher, you must be extremely careful about:

- What you are writing.
- What the findings tell you.
- The interpretations of your findings.

It's about **being honest** and ensuring the data sets you use in assessments or real work are **accurate and verifiable**. Do not get information from the internet indiscriminately. Ensure you are using **trustworthy and verifiable sources**; otherwise, your findings will be misleading.

Summary and Invitation for Discussion

So, these are the characteristics of research and its different types. So far, can anyone relate to any of the things we've discussed? Or does anyone want to share an experience, make a comment, or give an example where research has been used unethically? Don't be shy.

> 🧑‍🏫 Lecturer: Share your experiences, your thoughts. This is how we learn together. Dongfang, maybe you can share your experiences?

研究特征：数据完整性与伦理操守

现阶段，你们中的一些人可能会觉得这非常理论化，或者难以将其与研究的具体实践步骤联系起来。但别担心。在接下来的课程中，随着我们收集数据、分析数据和回答问题，这些概念会变得更加清晰。

一个基本特征是：**研究必须使用精确的数据**。例如，如果要收集数据来了解UB大学中承受压力的学生数量，数据必须是准确的。如果你使用的数据不准确，你的研究结果就会不准确，从而导致基于错误信息的决策失误。因此，数据收集和验证必须精确。

如果你使用来自其他来源的数据，你需要确保或验证这些数据是可靠的。在你的评估中，通过选择**同行评审的出版物**来实现这一点。你还必须保持**可信度**，并谨慎对待**诚信**。

伦理诚信的重要性：一个案例研究

我给你们举个例子。在进行研究时，你会发现原住民群体，比如毛利社区，对于分享信息非常严格。你不能直接接触一个毛利社区进行研究。你必须通过我们所说的**毛利顾问**——一个了解你并了解社区的受信任的人——来建立信任，然后才能进行访谈或收集数据。

这一要求源于非常合理的历史原因。历史上，这些社区曾遭受滥用。关于他们收集的数据在发表时，往往不能反映他们实际所说的情况。

例如，考虑一下毛利和太平洋社区肥胖率较高这一说法。一个简单化、不道德的结论可能是他们没有遵循健康的生活方式或运动不足。这是不科学且不诚实的。需要进行适当的研究来理解原因，这些原因可能包括：

- 遗传因素。
- 压力。
- 社会经济地位（例如，无力负担健康食品，而快餐更便宜）。

当你获取数据、分析数据，然后未能正确呈现时，这就表现出**缺乏诚实和缺乏诚信**。作为一名研究者，你必须极其谨慎地对待：

- 你所写的内容。
- 研究结果告诉你的信息。
- 你对研究结果的解释。

这关乎**诚实**，并确保你在评估或实际工作中使用的数据集是**准确且可验证的**。不要随意从互联网上获取信息。确保你使用的是**可信且可验证的来源**；否则，你的研究结果将具有误导性。

总结与讨论邀请

那么，这些就是研究的特征及其不同类型。到目前为止，有人能联系到我们讨论过的任何内容吗？或者有没有人想分享一次经历、发表评论，或举出一个研究被不道德使用的例子？不要害羞。

> 🧑‍🏫 讲师：分享你的经验，你的想法。我们就是这样一起学习的。东方，也许你可以分享一下你的经验？

总结与讨论邀请

那么，这些就是研究的特征及其不同类型。到目前为止，有人能联系到我们讨论过的任何内容吗？或者有没有人想分享一次经历、发表评论，或举出一个研究被不道德使用的例子？不要害羞。

> 🧑‍🏫 讲师：分享你的经验，你的想法。我们就是这样一起学习的。东方，也许你可以分享一下你的经验？

> 🙋‍♂️ 学生（东方）：教授您好。我听到您谈到了诚信。实际上我不是想谈诚信，我想谈谈论证。我的方法是，你应该让你的**主张**清晰，然后找到你的**数据**作为依据。接着你需要提供你的**理由**和**支持**，来用数据支撑你的主张。除此之外，你还需要找到一些**反驳**。完成这些工作后，我们可以将主张限定为一个更精确的结论。这就是我脑海中关于研究的一些想法。

> 🧑‍🏫 讲师：正确，正确。是的，我明白你的意思。你告诉我们你提出主张的步骤：你必须收集信息，收集文件，你必须基于一些可验证的东西提出主张。我理解得对吗？

> 🙋‍♂️ 学生（东方）：是的，是的。

> 🧑‍🏫 讲师：是的，即使在研究中，你也可以从研究问题开始，或者从一个**假设**开始。我们会学习这些内容。假设意味着一种**假定**。例如，你可以假定一个假设：在新西兰，男性的收入高于女性。这就是一个假设。但只要你没有进行研究，那只是一个假定。然后你去收集

数据，看看从哪里获取资源，分析你的数据。之后，基于此，数据要么支持你的假定，要么不支持。但归根结底，当你提出一个主张，一个科学主张时，它必须是**可验证的**，并且必须是**诚实的**。

🙋 学生（东方）：是的，我明白了。我发现数据的质量非常重要，获取正确、精确的数据可能需要花费很多时间和金钱。

👨‍🏫 讲师：正确。正确。对的。

课间休息与后续内容预告

在我们深入探讨研究类型之前，或者也许我们可以先完成这部分，然后我会给大家休息时间。通常我们休息大约15分钟，但今天因为我们在家里比较舒适，我会给大家大约5分钟，然后我们可以继续。

在我们休息之前，我简要地告诉你们我们回来后将要看的研究类型：

- 我们可以有**相关性研究**。
- 我们可以有**理论研究**。
- 我们可以有**应用研究**。

当我们学习这些内容时，不要认为你需要死记硬背。你不需要。我们现在的阶段不是死记硬背，而是**理解概念**。然后，当你需要做研究时，你就会明白，好吧，那是哪种类型的研究？我是在探索吗？我是在试图找出两个事物之间的关系吗？等等。

对于**相关性研究**，它识别两个或更多**变量**之间的关系。我现在使用的术语开始有点专业了。所以如果你觉得有点跟不上，请随时打断我。

好的。那么，我可以做研究来理解，这个和那个之间是否存在关系？例如，**睡眠不足**和**学业表现**之间是否存在关系？我想研究这个。

Summary and Discussion Invitation

So, these are the characteristics of research and its different types. So far, can anyone relate to any of the things we've discussed? Or does anyone want to share an experience, make a comment, or give an example where research has been used unethically? Don't be shy.

👨 🏫 Lecturer: Share your experiences, your thoughts. This is how we learn together. Dongfang, maybe you can share your experience with us?

🙋‍♂️ Student (Dongfang): Hello Professor. I heard you talk about integrity. Actually, I'm not trying to talk about integrity. I want to talk about the argument. That's what I'm trying to do —to put forward a valid argument. My method is that you should make your **claim** clear, then you would find your **data** as your ground. Then you find you need to give your **warrant** and **backing** to use the data to support your claim. Besides this, you need to find some **rebuttal**. After doing this work, we can qualify this claim to a more precise conclusion. This was some of the research in my mind.

👨‍🏫 Lecturer: Correct, correct. Yeah, I see your point. You're telling us the steps you take to make a claim: you have to collect your information, you have to collect your documents, you have to make a claim based on something verifiable. Did I get you right?

🙋‍♂️ Student (Dongfang): Yeah, yeah.

👨‍🏫 Lecturer: Yeah, so even in research, you can start with a research question or you can start with a **hypothesis**. We learn about these things. A hypothesis means an **assumption**. So you can assume, for example, your hypothesis would be: men earn more income than women in New Zealand. That would be a hypothesis. But so long as you didn't do your research, that's just an assumption. Then you go collect your data, see where you're going to get your resources from, analyze your data. Then based on that, it supports your assumption or not. But at the end of the day, when you make a claim, a scientific claim, it has to be **verifiable** and it has to be **honest**.

🙋‍♂️ Student (Dongfang): Yeah, I see. I find that the quality of the data is very important, and it may cost them much time and money to get the right, precise data.

👨‍🏫 Lecturer: Correct. Correct. Right.

Break Announcement and Preview of Upcoming Content

Before we delve into the types of research, or maybe we can finish this one, and then I'll give you a break. Usually, we take around 15 minutes, but because we're at the comfort of our homes today, I'll give you around five minutes, and then we can continue.

Just before we go for our break, I'll briefly tell you about the types of research we're going to look into when we come back:

- We can have **correlational research**.
- We can have **theoretical research**.
- We can have **applied research**.

When we learn about these things, don't think you're supposed to learn them by heart. You don't. We're not at a stage where we learn things by heart. It's about just **understanding the concepts**. Then, when the time comes for you to do your research, you'll figure out, okay, what type of research is that? Am I exploring? Am I trying to find a relationship between two things? And so on.

For **correlational research**, it identifies relationships between two or more **variables**. The terminology I'm using now is getting a little bit professional. So if you feel like you're a little bit lost, feel free to stop me.

Okay. So I can do research to understand: is there a relationship between this and that or not? For example, is there a relationship between **lack of sleep** and **academic performance**? I want to research this.

Correlational Research

Correlational research identifies relationships between two or more variables. For example, one might investigate:

- Is there a relationship between **lack of sleep** and **academic performance**?
- Is there a relationship between time spent on health apps and physical activity levels?
- Is there a relationship between screen time and mental health?

The core purpose is to find if a relationship exists between variables **without assuming causation**. Causation means one variable directly causes a change in another. For instance, finding a relationship between less screen time and better academic performance does not prove that reducing screen time *causes* improved grades; other factors could be involved.

> 🙋 Student: If we don't find any correlation between two variables, can we still call it correlation research? 🧑‍🏫 Lecturer: Yes, we can, because it identifies relationships between variables without assuming causation.

Theoretical vs. Applied Research

We can also conduct **theoretical research**, which advances knowledge by exploring theories and concepts without immediate practical application. Examples include:

- Investigating information behavior models to understand how patients seek online health advice.

- Exploring philosophical or theological concepts.

In contrast, **applied research** aims to solve a real-world problem directly. An example is testing a **machine learning algorithm** to improve diagnostic accuracy in radiology.

Most projects in fields like computer science involve these concepts: correlational, theoretical, or applied research. Towards the end of the course, I will introduce specific research types useful for capstone projects, such as **Design Science** (common in computer science and business analytics) and **Delphi studies** (for aggregating expert opinions).

相关性研究

相关性研究旨在识别两个或多个变量之间的关系。例如，可以研究：

- **睡眠不足**与**学业表现**之间是否存在关系？
- 使用健康应用的时间与身体活动水平之间是否存在关系？
- 屏幕使用时间与心理健康之间是否存在关系？

其核心目的是发现变量之间是否存在关系，**而不假定因果关系**。因果关系意味着一个变量直接导致另一个变量的变化。例如，发现减少屏幕时间与更好的学业成绩之间存在关系，并不能证明减少屏幕时间*导致*了成绩提高；可能涉及其他因素。

> 🙋‍♂️ 学生：如果我们没有发现两个变量之间存在任何相关性，还能称之为相关性研究吗？
> 🧑‍🏫 讲师：是的，可以。因为它是在不假定因果关系的前提下识别变量之间的关系。

理论与应用研究

我们还可以进行**理论研究**，这类研究通过探索理论和概念来推进知识发展，但没有直接的实际应用。例子包括：

- 调查信息行为模型，以了解患者如何在线寻求健康建议。
- 探索哲学或神学概念。

相比之下，**应用研究**旨在直接解决现实世界的问题。一个例子是测试**机器学习算法**以提高放射学诊断的准确性。

在计算机科学等领域，大多数项目都涉及这些概念：相关性研究、理论研究或应用研究。在课程接近尾声时，我将介绍对毕业设计项目有用的特定研究类型，例如**设计科学**（在计算机科学

和商业分析中常见）和**德尔菲研究**（用于汇总专家意见）。

Research Types (Continued)

Towards the end of the course, I will introduce two specific research types that might be useful for your capstone projects. One is **design science**, which is very common in fields like computer science and business analytics. Another is **Delphi studies**, which is used for aggregating expert opinions. Don't worry about these details now; we are still exploring what this research methods course is about.

Let's take a five-minute break. It's now 38 minutes past the hour. Let's meet again at 45 minutes past. See you in about six or seven minutes.

Exploratory, Descriptive, and Diagnostic Research

Exploratory Research

This type of research aims to provide familiarity with a phenomenon or achieve new insights. In simple terms, it's about exploring areas we don't fully understand. For example:

- Exploring the experiences of international students in New Zealand.
- Conducting initial studies on emerging technologies or new market trends.

We engage in exploratory research when we lack substantial information and are trying to generate new ideas or hypotheses about an uncharted area.

Descriptive Research

This involves accurately portraying the characteristics of a specific individual, situation, or group. It's essentially about describing what is happening. Examples include:

- Demographic studies.
- Surveys on consumer behavior.

For instance, a study might describe the shopping behavior of people in Auckland or report that "67% of students are satisfied with their experiences in New Zealand." The goal is to provide a detailed description to understand a phenomenon, such as students' well-being, without manipulating any variables.

Diagnostic Research

This type of research focuses on identifying the causes or reasons behind an event, or its association with other factors. An example is investigating the prevalence of a disease linked to certain risk factors.

For instance, if there is a high rate of a specific cancer in a certain region, diagnostic research would aim to understand if specific risk factors are associated with it. The goal is to diagnose the underlying reasons for an observed phenomenon.

Full Chinese Translation

研究类型（续）

在本课程接近尾声时，我将介绍两种可能对你们的毕业设计项目有用的具体研究类型。一种是**设计科学**，这在计算机科学和商业分析等领域非常常见。另一种是**德尔菲研究**，用于汇总专家意见。现在不必担心这些细节；我们仍在探索这门研究方法课程的内容。

我们休息五分钟。现在是整点过38分。我们45分时再见面。大约六、七分钟后见。

探索性、描述性与诊断性研究

探索性研究

这类研究旨在提供对某一现象的熟悉度或获得新的见解。简单来说，就是探索我们尚未完全理解的领域。例如：

- 探索国际学生在新西兰的经历。
- 对新兴技术或新市场趋势进行初步研究。

当我们缺乏足够信息，并试图对一个未知领域产生新想法或假设时，就会进行探索性研究。

描述性研究

这涉及准确描述特定个体、情况或群体的特征。本质上就是描述正在发生的事情。例子包括：

- 人口统计学研究。
- 消费者行为调查。

例如，一项研究可能描述奥克兰人的购物行为，或者报告"67%的学生对新西兰的经历感到满意"。其目标是提供详细描述以理解某种现象（如学生的福祉），而不操纵任何变量。

诊断性研究

这类研究侧重于识别事件背后的原因或理由，或其与其他因素的关联。一个例子是调查与某些风险因素相关的疾病患病率。

例如，如果某个地区某种特定癌症的发病率很高，诊断性研究将旨在了解是否存在与之相关的特定风险因素。其目标是诊断所观察现象的根本原因。

Diagnostic Research

An example of diagnostic research is investigating the prevalence of a disease linked to certain risk factors. The goal is to diagnose the reasons behind an observed phenomenon. For instance, if there is a high rate of a specific type of cancer in a certain region, the research aims to understand if there are specific risk factors associated with it. This type of research focuses on finding the cause of an event or its association with another factor, essentially looking for patterns or correlations between variables. It is also somewhat exploratory in nature, as it seeks to understand relationships between variables.

Hypothesis Testing Research

This type of research involves testing a hypothesis about a causal relationship between variables. An assumption remains just an assumption unless it is supported by evidence, which is the core of evidence-based research. Hypothesis testing aims to confirm or invalidate a proposed relationship.

Examples include clinical trials or experiments testing the effect of new teaching methods on student performance, or the impact of new drugs. For instance, to understand the impact of blended learning on student performance, one might start with the null hypothesis that there is no difference between blended and in-person learning.

🙋 Student: Does that make sense? 🧑‍🏫 Lecturer: Yes, ma'am. Cool.

The process involves collecting data (e.g., comparing grades and circumstances of online vs. in-person students), analyzing it, and reaching a conclusion. The finding might show that face-to-face students have higher grades, or the opposite could be true depending on various factors. The key point is that without data, a claim remains an assumption. Hypothesis testing is about determining whether an assumption is supported by evidence, and we will discuss this further in the context of quantitative research. It serves to confirm or invalidate a proposed relationship or to understand the 'why' and 'how' aspects.

Exercise Introduction

Now, I will give you a five-minute exercise. You will go through a set of scenarios and categorize each based on whether it is **exploratory**, **descriptive**, **diagnostic**, or **hypothesis testing**. You can always refer to the slides on Blackboard for guidance.

Just a quick reminder:

- **Exploratory**: Investigating a new, uncharted area.
- **Descriptive**: Describing what is happening.
- **Diagnostic**: Finding the cause or reason (diagnosing).
- **Hypothesis Testing**: Testing an assumption.

诊断性研究

诊断性研究的一个例子是调查与某些风险因素相关的疾病患病率。其目标是诊断所观察现象背后的原因。例如，如果某个地区某种特定癌症的发病率很高，该研究旨在了解是否存在与之相关的特定风险因素。这类研究侧重于寻找事件的原因或其与另一因素的关联，本质上是寻找变量之间的模式或相关性。它也具有一定的探索性，因为它试图理解变量之间的关系。

假设检验研究

这类研究涉及检验关于变量间因果关系的假设。一个假设除非有证据支持，否则就只是一个假设，这是循证研究的核心。假设检验旨在确认或否定一个提出的关系。

例子包括临床试验或测试新教学方法对学生表现影响的实验，或测试新药的影响。例如，为了理解混合式学习对学生表现的影响，可以从零假设开始，即混合式学习与面对面学习没有差异。

🙋 学生：这清楚吗？ 🧑‍🏫 讲师：是的，女士。很好。

这个过程包括收集数据（例如，比较在线学生与面对面学生的成绩和情况），分析数据，并得出结论。结果可能显示面对面学习的学生成绩更高，或者根据各种因素，相反的情况也可能成立。关键点在于，没有数据支持，一个主张就只是一个假设。假设检验是关于确定一个假设是否有证据支持，我们将在定量研究的背景下进一步讨论这一点。它用于确认或否定一个提出的关系，或理解“为什么”和“如何”的方面。

练习介绍

现在，我将给你们一个五分钟的练习。你们将浏览一组场景，并根据每个场景是**探索性**、**描述性**、**诊断性**还是**假设检验**来进行分类。你们可以随时参考Blackboard上的幻灯片作为指导。

快速提醒：

- **探索性**：调查一个新的、未知的领域。
- **描述性**：描述正在发生的事情。
- **诊断性**：寻找原因或理由（诊断）。
- **假设检验**：检验一个假设。

Exercise: Categorizing Research Scenarios

We will now begin a five-minute exercise. You will go through a set of scenarios and categorize each one as **exploratory**, **descriptive**, **diagnostic**, or **hypothesis testing**. You can always refer to the slides on Blackboard for guidance.

A quick reminder of the categories:

- **Exploratory**: Investigating a new, uncharted area.
- **Descriptive**: Describing what is happening.
- **Diagnostic**: Finding the cause or reason (often used in clinical/healthcare settings).
- **Hypothesis Testing**: Testing a specific assumption.

I will give you about five minutes to work through them. When I return, we will discuss your findings.

🙋 Student: Sorry, I don't have permission to access the course menu on Blackboard. Could you please share the slides in the chat or channel so I can download them? 🧑‍🏫

Lecturer: The slides, you mean? 🙋 Student: Yes, because I don't have access on Blackboard. I think it's a permission issue. 👨‍🏫 Lecturer: Okay, you can get in touch with Dr. Arun. Meanwhile, I'll minimize my screen and post them on Teams as you work through the exercise.

Reviewing the Scenarios

Let's go through these scenarios together.

Scenario 1: A company wants to understand why employees are quitting at a high rate. What kind of research is this?

- One answer was "Descriptive."
- Another perspective is that it could be **exploratory**, as they are exploring the reasons.
- It also has **diagnostic** elements (finding the cause), but the term "diagnostic" is typically used in clinical or healthcare settings rather than business informatics.

Scenario 2: A marketing firm surveys 5,000 consumers to understand their shopping behavior during the holidays.

- The consensus is **descriptive**. It is describing current behavior.

Scenario 3: A hospital studies the relationship between air pollution levels and the number of asthma cases.

- Some suggested "correlation."
- In the context provided, this is **diagnostic**. It is investigating a potential cause-and-effect relationship within a healthcare setting. It's important to note that you will find elements of each type in others. The main goal now is to familiarize ourselves with the terms.

Scenario 4: A university tests whether online learning improves student performance compared to traditional classrooms.

- This is **hypothesis testing**. You start with a null hypothesis (e.g., there is no difference) and test whether the data supports your hypothesis.
- It's good to hear other suggestions, as it shows you can see the relationships and overlaps between the different research types.

Scenario 5: A tech startup is exploring how artificial intelligence can enhance customer service.

- This is **exploratory**. The startup is investigating a new, uncharted application area.

练习：对研究场景进行分类

我们现在开始一个五分钟的练习。你们将浏览一组场景，并根据每个场景是**探索性**、**描述性**、**诊断性**还是**假设检验**来进行分类。你们可以随时参考Blackboard上的幻灯片作为指导。

各类别的快速提醒：

- **探索性**：调查一个新的、未知的领域。
- **描述性**：描述正在发生的事情。
- **诊断性**：寻找原因或理由（常用于临床/医疗保健环境）。
- **假设检验**：检验一个特定的假设。

我将给你们大约五分钟的时间来完成。等我回来，我们将讨论你们的发现。

> 🙋‍♂️ 学生：抱歉，我没有权限访问Blackboard上的课程菜单。您能否在聊天或频道中分享幻灯片，以便我下载？ 👨‍🏫 讲师：你是指幻灯片吗？ 🙋‍♂️ 学生：是的，因为我在Blackboard上没有访问权限。我想是权限问题。 👨‍🏫 讲师：好的，你可以联系Arun博士。同时，我会最小化我的屏幕，并在你们练习时将它们发布到Teams上。

回顾场景

让我们一起回顾这些场景。

场景1： 一家公司想了解为什么员工离职率很高。这是什么类型的研究？

- 一个答案是“描述性”。
- 另一个观点是，这可能是**探索性**的，因为他们正在探索原因。
- 它也具有**诊断性**元素（寻找原因），但“诊断性”一词通常用于临床或医疗保健环境，而不是商业信息学。

场景2： 一家营销公司调查了5000名消费者，以了解他们在假日期间的购物行为。

- 共识是**描述性**。它描述的是当前的行为。

场景3： 一家医院研究空气污染水平与哮喘病例数量之间的关系。

- 有人建议是“相关性”。
- 在所提供的上下文中，这是**诊断性**的。它是在医疗保健环境中调查潜在的因果关系。需要注意的是，你会在其他类型中发现每种类型的元素。目前的主要目标是熟悉这些术语。

场景4： 一所大学测试在线学习是否比传统课堂更能提高学生的表现。

- 这是**假设检验**。你从一个零假设（例如，没有差异）开始，然后检验数据是否支持你的假设。
- 听到其他建议是很好的，因为这表明你能看到不同类型研究之间的关系和重叠。

场景5： 一家科技初创公司正在探索人工智能如何提升客户服务。

- 这是**探索性**的。这家初创公司正在调查一个新的、未知的应用领域。

Scenario 5: Tech Startup & AI in Customer Service

This is **exploratory** research. The startup is investigating a new, uncharted application area. There is no clear hypothesis at the outset; the goal is to understand and explore a domain we currently do not fully comprehend.

> 🙋‍♂️ Student: Discriminatory. 👨‍🏫 Lecturer: I think it's exploratory. Why do you think it's exploratory? 🙋‍♂️ Student: Exploratory. 👨‍🏫 Lecturer: Exploratory. There is no clear hypothesis. Yeah, it's trying to understand or to explore uncharted areas. We don't understand this.

Research Type Overlaps & Determining Factors

The lecturer emphasizes that a single study can sometimes be viewed from multiple perspectives (e.g., descriptive or hypothesis testing), depending on the specific **research question**. This highlights the importance of clarity and precision in research design.

> 👨‍🏫 Lecturer: That's why I told you that you will find some studies that could be looked at from a descriptive, sometimes from a hypothesis testing, but it depends on what is the research question.

Further Scenario Examples

- **Scenario 6:** A sociologist conducts a national survey to measure public opinion on climate change.
  - This is **descriptive** research.
- **Scenario 7:** A healthcare study investigates whether a new diet plan reduces the risk of heart disease.
  - This could be considered **diagnostic** research. The lecturer notes it could also be seen as **hypothesis testing**, as it's testing a specific claim. The distinction can be subtle initially.
- **Scenario 8:** Researchers analyze hospital records to determine if certain age groups are more vulnerable to COVID-19 complications.
  - This is **diagnostic** research.

The Central Role of the Research Question

The key to determining the appropriate research type always **boils down to the research question**. This is the most important step in research. The process begins by developing a **well-defined research question**.

For **hypothesis testing**, this involves:

1. Starting with a **null hypothesis (H0)**, which typically states there is no relationship or no effect.
2. Defining an **alternative hypothesis**, which states there is a relationship or effect.
3. Collecting and analyzing data to see if you can support the null hypothesis.
   - If you **can support H0**, you conclude there is no relationship.
   - If you **cannot support H0**, you conclude there is a relationship.

Introduction to Qualitative vs. Quantitative Methods

Research methods are broadly categorized into **qualitative** and **quantitative** approaches.

- **Qualitative** research often deals with words, meanings, and in-depth understanding.
- **Quantitative** research deals with numbers, measurements, and statistical analysis.

The choice between them isn't always a matter of personal preference (e.g., being convinced by words vs. numbers). In practice, researchers are often constrained by factors like the **funding agency's** requirements, the nature of the research problem, and the available data.

完整中文翻译

场景五：科技初创公司与人工智能在客户服务中的应用

这是**探索性**研究。这家初创公司正在调查一个全新的、未知的应用领域。最初并没有明确的假设；其目标是理解并探索一个我们目前尚未完全理解的领域。

> 🙋‍♂️ 学生：判别式。 🧑‍🏫 讲师：我认为是探索性的。你为什么认为是探索性的？ 🙋‍♂️ 学生：探索性的。 🧑‍🏫 讲师：探索性的。没有明确的假设。是的，它试图理解或探索未知领域。我们对此并不了解。

研究类型的重叠与决定因素

讲师强调，根据具体的**研究问题**，一项研究有时可以从多个角度（例如，描述性或假设检验）来看待。这凸显了在研究设计中保持清晰和精确的重要性。

> 🧑‍🏫 讲师：这就是为什么我告诉你们，你们会发现有些研究可以从描述性的角度来看，有时也可以从假设检验的角度来看，但这取决于研究问题是什么。

更多场景示例

- **场景六：** 一位社会学家进行一项全国性调查，以衡量公众对气候变化的看法。
  - 这是**描述性**研究。
- **场景七：** 一项医疗保健研究调查新的饮食计划是否能降低患心脏病的风险。
  - 这可以被视为**诊断性**研究。讲师指出，由于它在检验一个特定的主张，也可以被看作是**假设检验**。这种区分在最初可能很微妙。
- **场景八：** 研究人员分析医院记录，以确定某些年龄组是否更容易出现 COVID-19 并发症。
  - 这是**诊断性**研究。

研究问题的核心作用

确定合适的研究类型，关键在于**研究问题**。这是研究中最重要的一步。整个过程始于提出一个**定义明确的研究问题**。

对于**假设检验**，这包括：

1. 从一个**零假设（H0）** 开始，该假设通常声明没有关系或没有影响。
2. 定义一个**备择假设**，声明存在关系或影响。
3. 收集和分析数据，以查看是否能支持零假设。
    - 如果**能支持 H0**，则得出结论认为没有关系。
    - 如果**不能支持 H0**，则得出结论认为存在关系。

定性方法与定量方法简介

研究方法大致分为**定性**和**定量**两种途径。

- **定性**研究通常处理文字、意义和深入的理解。
- **定量**研究处理数字、测量和统计分析。

两者之间的选择并不总是个人偏好的问题（例如，容易被文字说服还是被数字说服）。在实践中，研究人员常常受到诸如**资助机构**的要求、研究问题的性质以及可用数据等因素的限制。

定性研究与定量研究的区别与选择

定性和**定量**研究是两种主要的研究方法。选择哪种方法并不总是个人偏好问题，因为研究人员常常受到**资助机构**、可用资源等因素的限制。

- **定性研究**通常使用访谈、收集文字和人们的观点。它关注的是理解人们的经历、动机和态度，因此是**主观的**。
- **定量研究**则关乎测量和数字。它没有个人观点或经验的空间，是**客观的**。

核心区别：主观性与客观性

定性研究是主观的，因为它取决于个人的经历和感受。例如，研究国际学生在UB的经历，结果会因个人体验而异。 **定量研究**是客观的，因为它不受个人想法影响。例如，测试水是否在100摄氏度沸腾，这是一个可测量的客观事实。

数据类型与信息特征

- **定性研究**处理**非数值数据**（如文字），通过与被研究者的互动收集。它能提供丰富、深入的信息，帮助我们理解“为什么”。
- **定量研究**处理**数值数据**，测量变量（如心率、血压）。它更精确、客观，能告诉我们“发生了什么”，但通常无法提供深层解释。

混合方法

还存在**混合方法**研究，即结合使用定性和定量方法。可以先进行定量研究发现问题，再用定性研究深入探究原因；或者顺序相反。

> 🧑‍🏫 Lecturer: 大家明白区别了吗？如果我想了解“发生了什么”，我会选择定量研究。如果我想探究原因并深入挖掘，我会选择定性研究。

Qualitative vs. Quantitative Research: Differences and Selection

Qualitative and **quantitative** research are the two main methodological approaches. The choice is not always a matter of personal preference, as researchers are often constrained by factors such as **funding agencies** and available resources.

- **Qualitative research** typically uses interviews, collects words and people's opinions. It focuses on understanding people's experiences, motives, and attitudes, making it **subjective**.
- **Quantitative research** is about measurement and numbers. There is no room for personal opinion or experience; it is **objective**.

Core Distinction: Subjectivity vs. Objectivity

Qualitative research is subjective because it depends on individual experiences and feelings. For example, studying the experiences of international students at UB yields results that vary based on personal circumstances. **Quantitative research** is objective as it is not influenced by personal thoughts. For instance, testing whether water boils at 100 degrees Celsius is a measurable, objective fact.

Data Types and Information Characteristics

- **Qualitative research** deals with **non-numerical data** (e.g., words), collected through interactions with participants. It provides rich, in-depth information, helping us

understand the "why."
- **Quantitative research** deals with **numerical data**, measuring variables (e.g., heart rate, blood pressure). It is more precise and objective, telling us "what is happening," but often lacks deep explanatory power.

Mixed Methods

There is also **mixed methods** research, which combines qualitative and quantitative approaches. One might conduct quantitative research first to identify a phenomenon, then use qualitative research to explore the reasons in depth, or vice versa.

> 👨‍🏫 Lecturer: Can you see the difference? If I want to understand *what* is happening, I'll go for quantitative. If I want to understand the *reasons* and delve into it, I go for qualitative.

Research Methods: Qualitative, Quantitative, and Mixed Methods

Mixed Methods Research

Following the distinction between quantitative and qualitative approaches, there is also **mixed methods** research. In this approach, you can first conduct a quantitative study and then follow it with a qualitative one to understand *why* certain phenomena are occurring. Alternatively, you can do the opposite. Therefore, **mixed method research** involves using both qualitative and quantitative methods.

Interactive Check & Example Discussion

The lecturer conducted a quick check-in with students Cliff and Kiran to ensure everyone was following the concepts. The distinction between numerical (quantitative) and non-numerical (qualitative) data was confirmed as clear.

> 👨‍🏫 Lecturer: So instead of a hypothetical example, I'll ask some of you to give me an example of a research question that would be answered using qualitative or quantitative research. Think about any study you might want to conduct and tell me which approach you'd choose.

> 🙋‍♂️ Ravindo: It can be related to anything, right? I was thinking of a simple research question like, "Who would be a better president?" For research like that, when it comes to who would be a better president, it's mostly based on people's bias, emotions, and their experiences.

> 👨‍🏫 Lecturer: So, would you go for qualitative or quantitative?
>
> 🙋‍♂️ Ravindo: Yeah, it'll be qualitative. Because it doesn't deal with numbers; it deals with what people have experienced. So everyone's opinion is going to be different.
>
> 👨‍🏫 Lecturer: Nice one. Does anyone have any objection or comment about Ravindo's example? I'm assuming someone might say for the same question, they would want to do it quantitatively.
>
> 🙋‍♂️ Mohammed Sofyan: Because how I decide whether this person is a good president or not depends on the statistics I have. For example, how many people received good healthcare?
>
> 🙋‍♂️ Student: For this research question, if I reach out to people and get their opinion, then will it be categorized as quantitative because I'm getting a number from people's experiences?

Clarifying the Approach and Acknowledging Bias

The lecturer clarified the approach, focusing on the methodology rather than the specific research question.

> 👨‍🏫 Lecturer: If we go with Ravindo's qualitative approach, remember, I need a sample because I can't ask everyone in New Zealand. You must choose a **representative sample**. If I only ask people new to New Zealand, they might still be happy with the new environment and unaware of the challenges, which could lead to a **biased answer**.

Therefore, qualitative studies or approaches inherently contain some **bias**. This does not make them useless; they are still valuable. However, we must **acknowledge this bias**.

> 👨‍🏫 Lecturer: Don't worry about this now; we'll talk about it in more detail later. To summarize the methods:
>
> - If I ask people questions through **interviews**, it's **qualitative**.
> - If I ask questions using **surveys** and then **count** the number of people who are pro and anti, that's **quantitative**, as Mohammed Sofyan said.
> - Alternatively, I could also go back and perform proper **statistical analysis**.

研究方法：定性、定量与混合方法

混合方法研究

在区分了定量和定性方法之后，还存在**混合方法**研究。在这种方法中，你可以先进行定量研究，然后进行定性研究以理解某些现象发生的原因。或者，你也可以反过来进行。因此，**混合方法研究**涉及同时使用定性和定量方法。

互动检查与示例讨论

讲师与学生Cliff和Kiran进行了快速确认，以确保每个人都理解了这些概念。数值型（定量）和非数值型（定性）数据之间的区别被确认为是清晰的。

> 👨‍🏫 讲师：那么，与其举一个假设的例子，不如请你们中的一些人给我一个研究问题的例子，这个问题将使用定性或定量研究来回答。想想你可能想进行的任何研究，并告诉我你会选择哪种方法。
>
> 🙋‍♂️ Ravindo：可以涉及任何主题，对吧？我在想一个简单的研究问题，比如"谁会成为更好的总统？"对于这类研究，当涉及到谁会成为更好的总统时，这主要基于人们的偏见、情感和他们的经历。
>
> 👨‍🏫 讲师：那么，你会选择定性还是定量方法？
>
> 🙋‍♂️ Ravindo：是的，会是定性的。因为它不涉及数字；它涉及人们的经历。所以每个人的意见都会不同。
>
> 👨‍🏫 讲师：很好。有人对Ravindo的例子有异议或评论吗？我假设有人可能会说，对于同一个问题，他们想用定量方法来做。
>
> 🙋‍♂️ Mohammed Sofyan：因为我判断这个人是否是一个好总统，取决于我掌握的统计数据。例如，有多少人获得了良好的医疗保健？
>
> 🙋‍♂️ 学生：对于这个研究问题，如果我接触人们并获取他们的意见，那么这会因为我是从人们的经验中获取数字而被归类为定量研究吗？

澄清研究方法并承认偏见

讲师澄清了研究方法，侧重于方法论而非具体的研究问题。

> 👨‍🏫 讲师：如果我们采用Ravindo的定性方法，请记住，我需要一个样本，因为我无法询问新西兰的每一个人。你必须选择一个**代表性样本**。如果我只询问刚到新西兰的人，他们可能仍然对新环境感到满意，并且不了解其中的挑战，这可能导致**有偏见的答案**。

因此，定性研究或方法本身包含一些**偏见**。这并不会使它们变得无用；它们仍然有价值。然而，我们必须**承认这种偏见**。

> 👨 🏫 讲师：现在不用担心这个；我们稍后会详细讨论。总结一下这些方法：
>
> - 如果我通过**访谈**向人们提问，那就是**定性**的。
> - 如果我使用**调查问卷**提问，然后**统计**支持和反对的人数，那就是**定量**的，正如Mohammed Sofyan所说。
> - 或者，我也可以回头进行适当的**统计分析**。

Summary of Qualitative vs. Quantitative Methods

We must **acknowledge the bias**. To summarize these methods:

- If I ask people questions through **interviews**, it's **qualitative**.
- If I ask questions using **surveys** and then **count** the number of people for and against, it's **quantitative**, as Mohammed Sofyan said.
- Alternatively, I could also go back and perform proper **statistical analysis**, for example, by examining healthcare data like hospital waiting times or cancer treatment outcomes. The first step is to identify what I am measuring.

Choosing a Research Method

The same research question can be answered using either a qualitative or quantitative approach. The choice depends on several factors:

- **Resources**: Do you have the means to conduct qualitative or quantitative research?
- **Time**: What is your timeframe?
- **Preference & Justification**: Why are you choosing one method over the other?

This justification is crucial for assessments. When you justify your choice, you must also discuss the alternative method you did **not** select. You don't actually conduct the study both ways; instead, you argue that you chose, for example, a qualitative approach because it is better for specific reasons, while a quantitative approach would have certain other characteristics or be less feasible due to resource constraints.

> 👨 🏫 Lecturer: Did I lose you or you got me? 👨 🏫 Lecturer: So, to select one research method, we have to compare both and then conclude which is better for our specific case, explaining why we chose it.

Recap and the Importance of Specific Questions

Returning to the core distinction:

- If you **count numbers and statistics**, it's quantitative.
- If you explore **people's experiences and opinions**, it's qualitative.

The initial example question, "Who would make a better president?", is too vague. A good research question must be:

- **Specific** (e.g., which country? which political parties?)
- **Significant**
- **Measurable**

We will delve into developing good research questions later, as it is key to the first assessment.

Introduction to the Scientific Method

Research is a **journey**, not a single task. It begins with **observing a problem**.

定性研究与定量方法总结

我们必须**承认这种偏见**。总结一下这些方法：

- 如果我通过**访谈**向人们提问，那就是**定性**的。
- 如果我使用**调查问卷**提问，然后**统计**支持和反对的人数，那就是**定量**的，正如Mohammed Sofyan所说。
- 或者，我也可以回头进行适当的**统计分析**，例如，检查医疗数据，如医院候诊时间或癌症治疗结果。第一步是确定我要测量什么。

选择研究方法

同一个研究问题既可以用定性方法也可以用定量方法来回答。选择取决于以下几个因素：

- **资源**：你是否有进行定性或定量研究的手段？
- **时间**：你的时间框架是怎样的？
- **偏好与论证**：你为什么选择这种方法而不是另一种？

这种论证对于评估至关重要。当你论证你的选择时，你必须同时讨论你**没有**选择的替代方法。你实际上并不会用两种方式都进行研究；相反，你需要论证你选择了（例如）定性方法，因为它对于特定原因更优，而定量方法则会有某些其他特点，或者由于资源限制而可行性较低。

> 👨‍🏫 讲师：我跟丢你们了，还是你们跟上了？ 👨‍🏫 讲师：所以，要选择一种研究方法，我们必须比较两者，然后得出结论哪种更适合我们的具体情况，并解释我们选择它的原因。

回顾与明确问题的重要性

回到核心区别：

- 如果你**统计数字和数据**，那就是定量的。
- 如果你探索**人们的经验和观点**，那就是定性的。

最初的示例问题“谁会成为更好的总统？”过于模糊。一个好的研究问题必须是：

- **具体的**（例如，哪个国家？哪些政党？）
- **有意义的**
- **可测量的**

我们稍后将深入探讨如何提出好的研究问题，因为这是第一次评估的关键。

科学方法简介

研究是一个**旅程**，而非单一任务。它始于**观察一个问题**。

The Scientific Method: A Journey

Research is a **journey**, not a single task. It begins with **observing a problem or phenomenon**. You then ask a question. Think of it like someone noticing something, questioning it, developing a **hypothesis**, designing an experiment to test it, analyzing the data, drawing conclusions, and finally reporting the findings.

A crucial part of research is **reporting what you find**. This is how you add to existing knowledge. Writing papers or publications is essential; without this communication, it is not complete research.

Stages of the Scientific Method

This method is **scientific** because it is systematic, not haphazard, and aims to avoid or acknowledge bias. It is not merely about opinions. The key stages are:

1. **Observation**
2. **Formulating a research question**
3. **Developing a hypothesis** (Note: Not all studies require a formal hypothesis)
4. **Experimenting/Collecting data** to test the hypothesis
5. **Analyzing the data** (organizing it and making inferences)
6. **Reaching a conclusion**
7. **Communicating results** through papers, presentations, or reports

The outcome is **solutions or generalizations that advance knowledge**.

Research Ethics: A Foundational Principle

Research ethics is a critical, recurring theme. It refers to the **moral principles** governing a researcher's conduct before, during, and after a study.

Ethical obligations include:

- **Informed Consent:** Clearly informing participants about the study's time commitment, nature, and procedures.
- **Voluntary Participation:** Ensuring participants know they can withdraw at any time without penalty or harm.
- **Data Integrity:** It is **unethical** to analyze data and produce results that misrepresent what participants genuinely shared.

For example, when inviting someone for an interview, you must specify its duration, topics, and their right to withdraw. You cannot imply any negative consequences for non-participation.

完整中文翻译

科学方法：一段旅程

研究是一段**旅程**，而非单一任务。它始于**观察一个问题或现象**。然后你提出一个问题。可以想象为：有人注意到某事，产生疑问，提出一个**假设**，设计实验来检验它，分析数据，得出结论，并最终报告发现。

研究的一个关键部分是**报告你的发现**。这是你为现有知识添砖加瓦的方式。撰写论文或发表成果至关重要；没有这种交流，就不是完整的研究。

科学方法的阶段

这个方法之所以是**科学的**，是因为它是系统性的、非随意的，并且旨在避免或承认偏见。它不仅仅是关于观点。关键阶段包括：

1. **观察**
2. **提出研究问题**
3. **提出假设**（注意：并非所有研究都需要正式的假设）
4. **实验/收集数据**以检验假设
5. **分析数据**（组织数据并做出推断）
6. **得出结论**
7. **交流结果**，通过论文、演示或报告

其成果是**能够推动知识的解决方案或概括性结论**。

研究伦理：一项基本原则

研究伦理是一个关键且反复出现的主题。它指的是在研究之前、期间和之后**指导研究者行为的道德原则**。

伦理义务包括：

- **知情同意：** 向参与者清晰说明研究的时间投入、性质和流程。
- **自愿参与：** 确保参与者知道他们可以随时退出，且不会受到惩罚或伤害。
- **数据诚信：** 分析数据并产生**歪曲**参与者真实分享内容的结果是**不道德的**。

例如，邀请某人进行访谈时，你必须说明访谈的时长、主题以及他们退出的权利。你不能暗示不参与会带来任何负面后果。

Research Ethics and Integrity

Ethical Obligations to Participants

When conducting research involving human participants, specific ethical obligations must be clearly communicated and upheld.

- **Informed Consent:** Researchers must inform participants about the interview's duration, the topics to be discussed, and the nature of the questions.
- **Voluntary Participation & No Harm:** Participants must be assured that they can withdraw from the study at any time without penalty or harm. For example, their academic marks will not be affected if they choose not to share their information.
- **Withdrawal Condition:** This right to withdraw applies as long as their data has not yet been included in the formal analysis.

Safeguards for Sensitive Research

For studies on sensitive topics (e.g., mental health), additional safeguards are crucial. Researchers must consider the potential for questions to trigger emotional distress and have appropriate support mechanisms in place, such as access to psychological support. The goal is to ensure safety for both the researcher and the participant, creating a balanced and legally sound research relationship.

Definition and Scope of Research Ethics

Research ethics refers to the moral principles that govern a researcher's conduct before, during, and after a study. Historical abuses, such as unauthorized medical experiments on Black Americans, led to the establishment of strict regulations. In New Zealand, any research involving human participants must undergo formal **ethical approval** by an institution (e.g., a university) before it can begin. Researchers must submit their plans and proposed safeguards for review.

Ethics vs. Integrity

It is important to distinguish between ethics and integrity in research.

- **Ethics** are the moral principles guiding the research process itself.
- **Integrity** involves avoiding academic misconduct such as **plagiarism**, **fraud**, and **data fabrication**.

> 🧑‍🏫 Lecturer: I really hope no one goes through these experiences. Last term we had some unpleasant incidents with students regarding plagiarism and data fabrication. Please avoid these. Create your own work.

Core Purpose and Areas of Concern

The purpose of research ethics is to ensure that research does not cause harm to people, animals, or the environment and is conducted with integrity and fairness. Many contemporary issues (e.g., new nuclear technologies) raise ethical questions. Broad areas of ethical concern include:

1. Human and animal experimentation.
2. Environmental and societal impact.
3. Long-term implications for future generations.

Researchers must consider the impact of their work on future generations. An illustrative perspective comes from Māori, the indigenous people of New Zealand, whose worldview sees humans as part of the environment, not separate from it.

研究伦理与学术诚信

对参与者的伦理义务

在进行涉及人类参与者的研究时，必须清晰传达并恪守特定的伦理义务。

- **知情同意：** 研究者必须告知参与者访谈的时长、将要讨论的主题以及问题的性质。
- **自愿参与与无伤害：** 必须向参与者保证，他们可以随时退出研究，不会受到惩罚或伤害。例如，如果他们选择不分享信息，其学术成绩不会受到影响。
- **退出条件：** 只要他们的数据尚未被纳入正式分析，这项退出权利就适用。

敏感研究的保障措施

对于涉及敏感主题（例如心理健康）的研究，额外的保障措施至关重要。研究者必须考虑到问题可能引发情绪困扰，并准备好适当的支持机制，例如提供心理支持渠道。目标是确保研究者和参与者双方的安全，建立一种平衡且合法的研究关系。

研究伦理的定义与范围

研究伦理 指的是在研究之前、期间和之后指导研究者行为的道德原则。历史上的滥用行为，例如未经授权对美国黑人进行的医学实验，促使了严格法规的建立。在新西兰，任何涉及人类参与者的研究都必须先经过机构（例如大学）的正式**伦理审查批准**才能开始。研究者必须提交其研究计划和拟议的保障措施以供审查。

伦理与诚信之区别

区分研究中的伦理与诚信非常重要。

- **伦理** 是指导研究过程本身的道德原则。
- **诚信** 涉及避免学术不端行为，如**剽窃**、**欺诈**和**数据捏造**。

> 🧑‍🏫 讲师：我真的希望没有人经历这些。上学期我们遇到了一些涉及学生剽窃和数据捏造的不愉快事件。请避免这些行为。创作你自己的作品。

核心目的与关注领域

研究伦理的目的是确保研究不会对人员、动物或环境造成伤害，并以诚信和公平的方式进行。许多当代问题（例如新的核技术）引发了伦理质疑。广泛的伦理关注领域包括：

1. 人类与动物实验。
2. 环境与社会影响。
3. 对子孙后代的长期影响。

研究者必须考虑其工作对子孙后代的影响。一个具有启发性的视角来自新西兰原住民毛利人，他们的世界观认为人类是环境的一部分，而非与之分离。

毛利世界观与伦理考量

研究者必须考虑其工作对子孙后代的影响。一个具有启发性的视角来自新西兰原住民毛利人，他们的世界观认为人类是环境的一部分，而非与之分离。

当你参加会议做自我介绍时，你会习惯这样说（如果你在这里工作的话）：“这是我的名字，这是我居住地旁边的河流，它叫‘Pipiha’（可以查一下）。” 这表明他们不将自己与环境割裂开来。这在审视伦理问题时非常重要。

伦理准则与研究的可信度

诸如1964年《赫尔辛基宣言》等伦理准则，是在回应二战时期参与者被蓄意误导和伤害的丑闻后制定的。因此，研究的伦理方面不仅关乎保护，也关乎研究的**可信度**。如果不讨论伦理和文化方面，你的研究可能就不可信。

所有这些内容都将是你们在第一学期和第三学期的评估任务。

课程结构与研究应用

我将总结一下，这次课比平时稍短，但这只是一个介绍。第一学期的课程是基础课程。第三学期则是将基础理论或研究与实际项目相结合。

我们现在所讲的研究方法，将在第三学期的**顶点项目**中得到应用。

信息学中的不道德研究案例

以下是信息学中不道德研究的例子：

1. **市场研究中的数据操纵**：即数据捏造。如果你提供错误的数据，基于此数据做出的决策也将是错误的，这会误导公众。
2. **招聘中的偏见算法**：这是一个非常普遍的问题。例如，有时姓名、出身或其他特征可能被用于通过AI来排除或提拔某些人从事特定工作。这些都是信息学研究中糟糕的例子。
3. **医疗保健中未经同意的患者数据使用**：这会侵犯隐私并缺乏**知情同意**。

知情同意的核心要素

不要将任何人纳入任何未经其同意的研究。**知情同意**的概念意味着：

- 他们同意你所做的事情。
- 他们理解项目是什么。
- 他们同意你的做法，并且了解你将如何使用他们的数据。
- 他们知道你将把数据存储在哪里、存储多久，以及在数据使用目的完成后你将如何处置它。

例如，如果我收集数据，对Yirong进行访谈，却将其用于另一个目的或另一项研究，这是不道德的，并且可能产生法律后果。

伦理在行业与学术中的重要性

我所说的这些内容不仅是为了课程，对于你们理解行业工作和学术工作都至关重要。在研究中也必须思考伦理问题。

不要只选择能带来金钱收益的课题，要选择纳入代表性不足的群体，例如残疾人。这与软件工程密切相关，因为你可以为任何学科（如医疗保健或商业分析）设计软件。无论何时你设计某

个东西，你都会进行研究。

Māori Worldview and Ethical Considerations

Researchers must consider the impact of their work on future generations. An enlightening perspective comes from the indigenous Māori people of New Zealand, whose worldview holds that humans are part of the environment, not separate from it.

When you introduce yourself in a meeting, you'll get used to saying (if you work here): "This is my name, this is the river I live next to, it's called 'Pipiha' (look it up)." This shows they do not see themselves as separate from their environment. This is very important when examining ethical issues.

Ethical Codes and Research Credibility

Ethical codes such as the 1964 **Declaration of Helsinki** were developed in response to World War II-era scandals where participants were deliberately misled and harmed. Therefore, the ethical aspects of research are not only about protection but also about the **credibility** of the research. If you don't discuss ethical and cultural aspects, your research might not be credible.

All of these things will be part of your assessment tasks in Trimester One and Trimester Three.

Course Structure and Research Application

I'll conclude by saying this session is a bit shorter than usual, but it's just an introduction. Trimester One courses are foundational. Trimester Three combines foundational theory or research with practical projects.

The research methods we are discussing now will be applied in your **capstone projects** in Trimester Three.

Examples of Unethical Research in Informatics

Here are examples of unethical research in informatics:

1. **Data manipulation in market research**: This is data fabrication. If you provide wrong data, the decisions made based on this data will be wrong, misleading the public.
2. **Bias algorithms in hiring**: This is a very common problem. For example, sometimes names, origins, or other characteristics might be used by AI to exclude or promote certain people for specific jobs. These are bad examples of research in informatics.
3. **Unconsented use of patient data in healthcare**: This violates privacy and lacks **informed consent**.

Core Elements of Informed Consent

Do not include anyone in any research without their consent. The concept of **informed consent** means:

- They agree to what you are doing.
- They understand what the project is.
- They agree to your actions and understand how you will use their data.
- They know where you will store the data, for how long, and how you will dispose of it after its purpose is fulfilled.

For example, if I collect data, conduct an interview with Yirong, and then use it for another purpose or another study, this is unethical and has legal implications.

The Importance of Ethics in Industry and Academia

What I'm saying is not only for the course but is crucial for your understanding of both industry and academic work. Always think about ethics in research.

Don't just choose topics that bring money; choose to include underrepresented groups, such as people with disabilities. This is highly relevant to software engineering because you can design software for any discipline, like healthcare or business analytics. Whenever you design something, you will be conducting research.

Research Ethics and Session Conclusion

Ethical Considerations in Research and Software Engineering

Also consider **ethics in research**. Do not choose projects solely for monetary gain. Instead, prioritize inclusivity by involving underrepresented groups, such as people with disabilities.

This is highly relevant to software engineering because you can design software for any discipline, like healthcare or business analytics. Whenever you design something, you will be conducting research. Be inclusive:

- Include those who are underrepresented.
- Include people with physical or mental disabilities.
- Include people beyond frequently seen age groups, such as elderly populations.

All of these are examples of ethical aspects we need to focus on.

Session Wrap-up and Administrative Matters

That concludes our main content a little earlier than usual. I hope this session gives you a good understanding of the course content and research in general. If you have any questions or thoughts about the course or research approaches, please let us know.

Before we finish, for those who joined late, please state your names for attendance.

> 🙋‍♂️ Student: Can you please add my name? Mahima Hakmim. 👨‍🏫 Lecturer: Mahima is present. Who else? 🙋‍♂️ Student: Anthony. 👨‍🏫 Lecturer: So far I have Anthony, Niraj... 🙋‍♂️ Student: Hi, ma'am. 👨‍🏫 Lecturer: Hi. So, we have Mahima, Anthony, Mohamed Sufyan. Who else? 🙋‍♂️ Student: Neeraj. 👨‍🏫 Lecturer: Neeraj. Okay. Just to make sure I haven't missed anyone, please type your names in the chat. *(Various students type names: Kiran, Nazmanesha, Sandali, Yasas, etc.)* 👨‍🏫 Lecturer: I can see all of them. I'll check the chat again later. 🙋‍♂️ Student: Hello, Riem. 👨‍🏫 Lecturer: Hi, darling. Just before we end, for everybody's benefit, if you have issues with your Blackboard access, please type your name or the issue in the chat so we can collate them.

Class Representative and Issue Escalation

> 👨‍🏫 Lecturer: Who is your class representative? 🙋‍♂️ Student: It's me. 👨‍🏫 Lecturer: Okay. For issues, as Darling mentioned, always communicate anything common to the whole class to him. He will then take it to the TA, Mohamed. If Mohamed is not the right person, he will take it to Arun. So, any issues with Blackboard, tell Darling. Darling will tell the tutor, and they will take it from there.

Final Remarks

If there are no other questions, we'll call it for today. Enjoy the rest of your weekend and stay safe. For those new to New Zealand, don't risk going out today due to the weather. If you have shopping to do, do it as soon as possible. Better to be prepared than to regret it later. Stay safe, take care of yourselves, and see you next week.

研究伦理与课程结束

研究与软件工程中的伦理考量

同时也要考虑**研究伦理**。不要仅仅为了金钱利益选择项目。相反，应优先考虑包容性，让 underrepresented 群体参与进来，例如残障人士。

这与软件工程高度相关，因为你可以为任何学科设计软件，比如医疗保健或商业分析。每当你设计某物时，你都将进行研究。务必保持包容：

- 包含那些未被充分代表的群体。
- 包含有身体或精神残障的人士。
- 包含超出常见年龄段的群体，例如老年人群。

所有这些都是我们需要关注的伦理方面的例子。

课程总结与行政事务

我们的主要内容比平时稍早一些结束。希望本次课程能让你对课程内容和研究有一个良好的整体理解。如果你对课程或研究方法有任何问题或想法，请告诉我们。

在结束之前，请迟到的同学报一下名字以便考勤。

> 🙋 学生：请加上我的名字，Mahima Hakmim。 👨‍🏫 讲师：Mahima 到了。还有谁？ 🙋 学生：Anthony。 👨‍🏫 讲师：目前有 Anthony, Niraj... 🙋 学生：老师好。 👨‍🏫 讲师：你好。那么，我们有 Mahima, Anthony, Mohamed Sufyan。还有谁？ 🙋 学生：Neeraj。 👨‍🏫 讲师：Neeraj。好的。为了确保我没有漏掉任何人，请在聊天框中输入你的名字。 *(多位学生输入名字：Kiran, Nazmanesha, Sandali, Yasas 等)* 👨‍🏫 讲师：我都能看到。稍后我会再检查一下聊天记录。 🙋 学生：你好，Riem。 👨‍🏫 讲师：你好，Darling。在结束之前，为了大家的利益，如果你在访问 Blackboard 时遇到问题，请在聊天框中输入你的名字或问题，以便我们汇总。

班级代表与问题反馈流程

🧑‍🏫 讲师：谁是你们的班级代表？ 🙋‍♂️ 学生：是我。 🧑‍🏫 讲师：好的。关于问题，正如 Darling 所说，任何涉及全班同学的事务都请先与他沟通。然后他会将问题传达给助教 Mohamed。如果 Mohamed 不是对接人，他会再传达给 Arun。所以，任何关于 Blackboard 的问题，告诉 Darling。Darling 会告知导师，然后他们会跟进处理。

最后提醒

如果没有其他问题，我们今天到此结束。祝大家周末愉快，注意安全。对于刚到新西兰的同学，由于天气原因，今天不要冒险外出。如果有购物需求，请尽快完成。有备无患总比事后后悔好。注意安全，照顾好自己，我们下周见。

Lecture Conclusion & Safety Reminder

If there are no other questions, we will conclude for today. Enjoy the rest of your weekend and stay safe.

For those who are new to New Zealand, do not risk going out today due to the weather. If you have any shopping to do, complete it as soon as possible. It is better to be prepared than to regret it later.

Take care of yourselves, stay safe, and I will see you all next week.

Summary & Recommended Resources

Summary

- **Admin Matters:** The session concludes with a strong emphasis on student safety, particularly for newcomers to New Zealand, advising against unnecessary travel due to adverse weather conditions.
- **Academic Concepts:** This segment is purely administrative, containing no new academic content related to Research Methods.

Recommended Level 9 Resources

As this segment contains no academic content, no specific resources are recommended. Please refer to the resource lists from previous academic content segments for materials on research methodologies, literature review, and academic writing.

完整中文翻译

讲座结束与安全提醒

如果没有其他问题，我们今天到此结束。祝大家周末愉快，注意安全。

对于刚到新西兰的同学，由于天气原因，今天不要冒险外出。如果有购物需求，请尽快完成。有备无患总比事后后悔好。

注意安全，照顾好自己，我们下周见。

总结与推荐资源

总结

- **行政事务：** 本次课程以强调学生安全结束，特别建议初到新西兰的学生因恶劣天气避免不必要的出行。
- **学术概念：** 此部分纯属行政事务，不包含与研究方**法**相关的新学术内容。

推荐 Level 9 资源

由于此片段不包含学术内容，因此不推荐特定资源。请参考之前学术内容片段中的资源列表，获取有关研究方法、文献综述和学术写作的材料。